

ASAD KHURSHID

Cybersecurity Analyst | BS Cybersecurity

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SUMMARY

Cybersecurity Analyst with hands-on experience across embedded security, cloud defense, and application security. Contracted by the National Center for Cyber Security to engineer an end-to-end encrypted voice communication system on embedded hardware. Proven ability to deliver across the full security stack - from Solidity smart contracts and AWS cloud infrastructure to cryptographic protocol implementation and AI-driven threat detection. ISC2 and AWS certified.

SKILLS

Security Operations: SIEM (Wazuh), IDS/IPS (Suricata), Log Analysis (Grafana), Incident Response, Root Cause Analysis

Cloud Security: AWS (ECS, RDS, CloudWatch, IAM), Docker, Kubernetes, Linux Administration

Vulnerability Assessment: Nmap, Wireshark, Vulnerability Classification, Remediation Planning

Network Defense: Firewalls (pfSense), Traffic Analysis, TCP/IP, DNS Security

Application Security: Secure Coding (OWASP), API Security, RBAC, Python, Node.js, Solidity

Embedded Systems: C++, Raspberry Pi, Bluetooth Low Energy (BLE), WebRTC, libsodium

WORK EXPERIENCE

National Center for Cyber Security - Embedded Security Developer (Contract) | Remote | Islamabad, Pakistan | 08/2025 - 05/2026

AHOM Limited - Backend Developer | Remote | 03/2025 - 05/2025

XPlay Studio - Full Stack Developer Intern | Remote | 10/2024 - 11/2024

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES) | Islamabad, Pakistan

Bachelor of Science in Cybersecurity | 08/2022 - 06/2026

Relevant Coursework: Network Security, Digital Forensics, Cloud Security, Ethical Hacking

PROJECTS

Blupher 1.0 - Encrypted Embedded Voice Communication System | Final Year Project | Funded by National Center for Cyber Security

- Real-time end-to-end encrypted voice over Raspberry Pi 4 using hybrid Bluetooth Low Energy and WebRTC relay; C++ pipeline (ALSA, Opus, XChaCha20-Poly1305) with X25519 ECDH key exchange, achieving 180-260 ms latency.

AI-Powered Vulnerability Detection and Remediation System | Academic Capstone

- Integrated Nmap scanning with a self-trained severity classifier and a local LLM (Qwen 3.0) to auto-generate remediation scripts, reducing incident response time.

TraceRoot - Incident Response and Root Cause Analysis Platform | AWS-Deployed SaaS

- Multi-tenant RCA tool on AWS ECS and RDS with JWT authentication, RBAC, Docker, SSL/TLS, and CloudWatch monitoring; includes Fishbone and 5-Why investigation workflows.

DeFi Freelance Marketplace - Blockchain Escrow Platform | Academic Capstone

- Full-stack decentralized application on Sepolia testnet: Solidity milestone-escrow contracts (Hardhat), NestJS and PostgreSQL REST API, Next.js 14 frontend with wagmi, RainbowKit, SIWE authentication, and IPFS deliverables.

CERTIFICATIONS

Certified in Cybersecurity (CC) - ISC2 - Credential ID: 2755065 | 08/2025 - 08/2028

AWS Academy Graduate - Cloud Foundations - AWS Training and Certification | 12/2025

Windows Forensics - Belkasoft | 03/2025

Linux Fundamentals - Coursera | 09/2022

ACTIVITIES

CyberSpace Legion CTF Competition - FAST-NUCES | 2024

- Competed in jeopardy-style challenges covering Cryptography, Web Exploitation, and Digital Forensics.